SG No. 127 D_{4h}^{5} $P4/mbm$ [tetragonal]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `4/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,7,8,9,10,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,5,6,11,12,13,14]`

Table 4: Wyckoff site: `2d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,7,8,9,10,15,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,5,6,11,12,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[5,6,7,8]`
3	$[0, 0, -z]$	`[9,10,11,12]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[13,14,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2,15,16]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[5,6,11,12]`
4	$\left[0, \frac{1}{2}, -z\right]$	`[7,8,9,10]`

Table 7: Wyckoff site: `4g`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	`[1,7,10,16]`
2	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	`[2,8,9,15]`
3	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	`[3,5,12,14]`
4	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	`[4,6,11,13]`

Table 8: Wyckoff site: `4h`, site symmetry: `m.2m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,7,10,16]`
2	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[2,8,9,15]`
3	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[3,5,12,14]`
4	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[4,6,11,13]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,10]`
2	$[-x,\, -y,\, 0]$	`[2,9]`
3	$[-y,\, x,\, 0]$	`[3,12]`
4	$[y,\, -x,\, 0]$	`[4,11]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	`[5,14]`
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[6,13]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	`[8,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `m..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,10]`
2	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[2,9]`
3	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[3,12]`
4	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[5,14]`
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[6,13]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[8,15]`

Table 11: Wyckoff site: `8k`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	`[1,16]`
2	$\left[-x,\, \frac{1}{2}-x,\, z\right]$	`[2,15]`

continued ...

Table 11

No.	position	mapping
3	$\left[\frac{1}{2}-x,\, x,\, z\right]$	[3,14]
4	$\left[x+\frac{1}{2},\, -x,\, z\right]$	[4,13]
5	$\left[\frac{1}{2}-x,\, x,\, -z\right]$	[5,12]
6	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	[6,11]
7	$\left[x,\, x+\frac{1}{2},\, -z\right]$	[7,10]
8	$\left[-x,\, \frac{1}{2}-x,\, -z\right]$	[8,9]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[-y,\, x,\, z]$	[3]
4	$[y,\, -x,\, z]$	[4]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[5]
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[6]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[7]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[8]
9	$[-x,\, -y,\, -z]$	[9]
10	$[x,\, y,\, -z]$	[10]
11	$[y,\, -x,\, -z]$	[11]
12	$[-y,\, x,\, -z]$	[12]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[16]